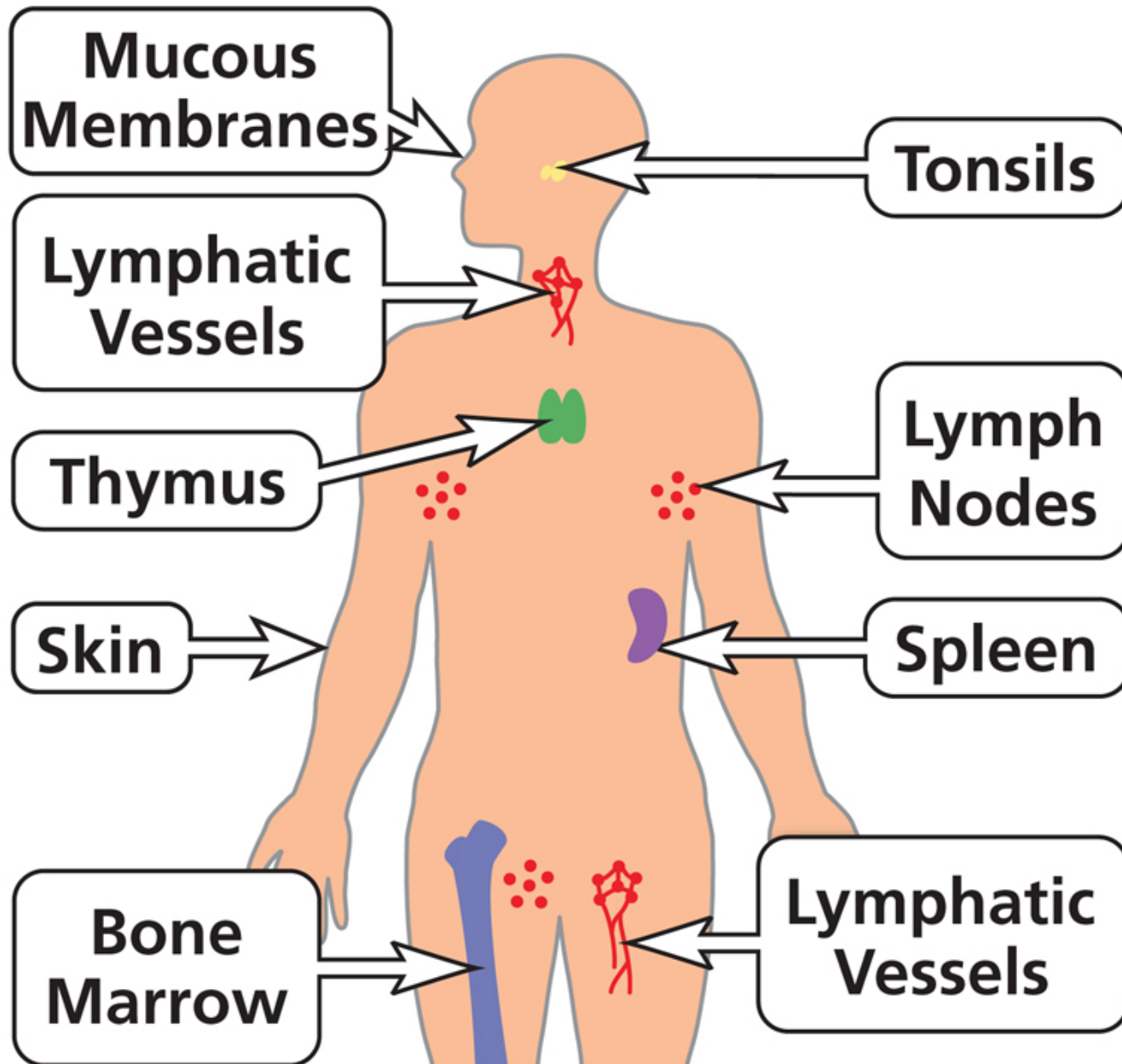




Histology of immune organs

Dr. Sándor Katz M.D.,Ph.D.

Immune System



Lymphoid organs

Primary lymphoid organs

- bone marrow
- thymus

Secondary lymphoid organs

- tonsils
- lymph nodes
- spleen

Framework of the lymphoid organs

Lymphoreticular connective tissue (tonsils, spleen and lymph nodes):

- reticular fibers
- reticulum cells
- immune cells

EXCEPT in the thymus:

Epithelioreticular cells form the structural framework.

Lymphatic tissue

Lymphocytes: main cellular constituents

Three major types:

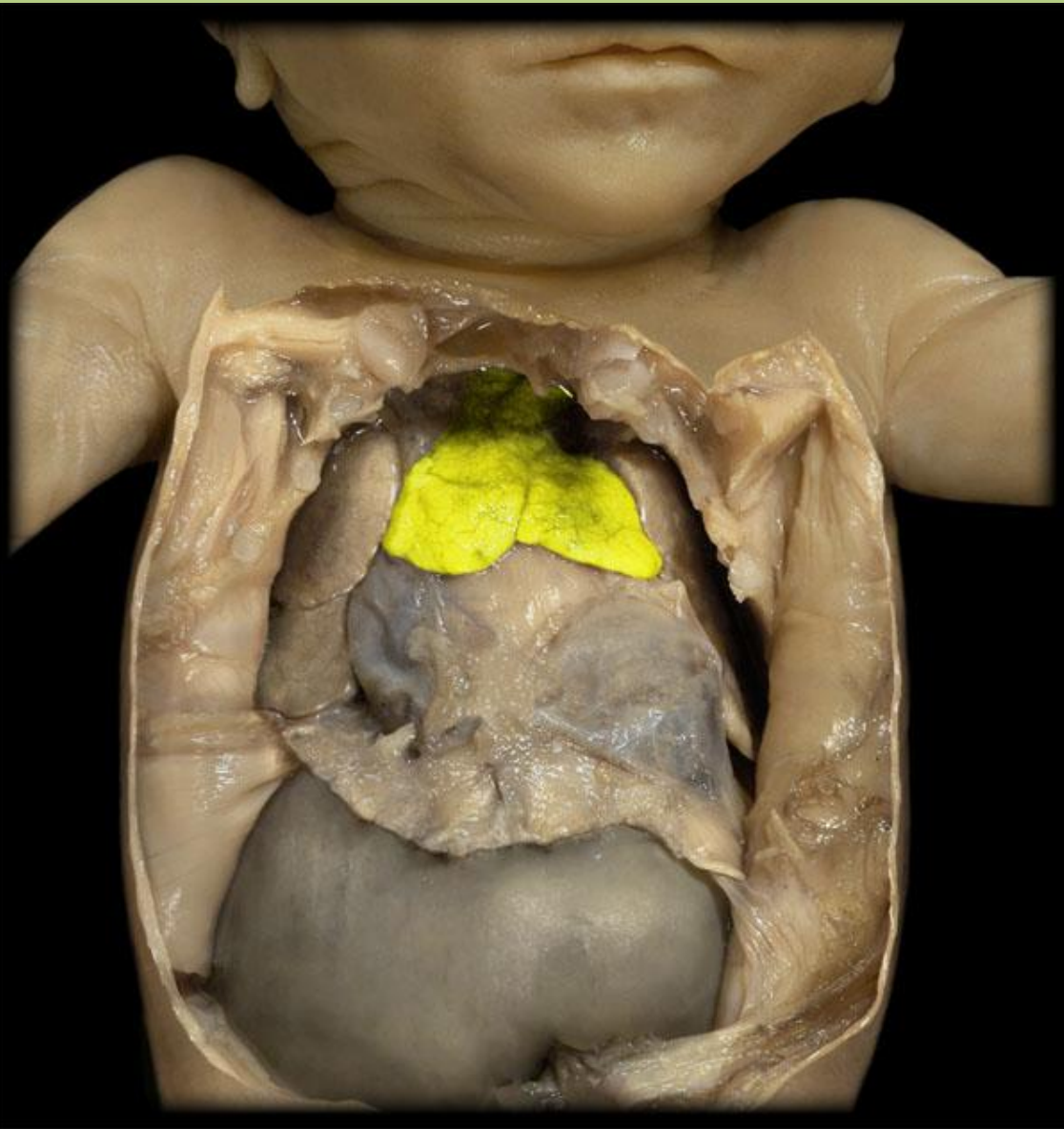
- **T cells:** cell-mediated immunity, secretion of cytokines
- **B cells:** humoral immunity, secretion of immunoglobulins
- **NK cells:** releasing of perforins (they create channels in the plasma membrane, initiating self-destruct)

Lymphatic tissue

Cells of the reticular meshwork:

- **Reticulum cells:** synthesis and secretion of type III collagen (reticular fibers)
- **Dendritic cells:** APCs (antigen presenting cells); they also monitor the local environment
- **Follicular dendritic cells:** thin hair-like cytoplasmic processes among the B cells in the germinal center; they help the debris removal and organize the lymphoid microarchitecture
- **Macrophages:** phagocytic and APCs

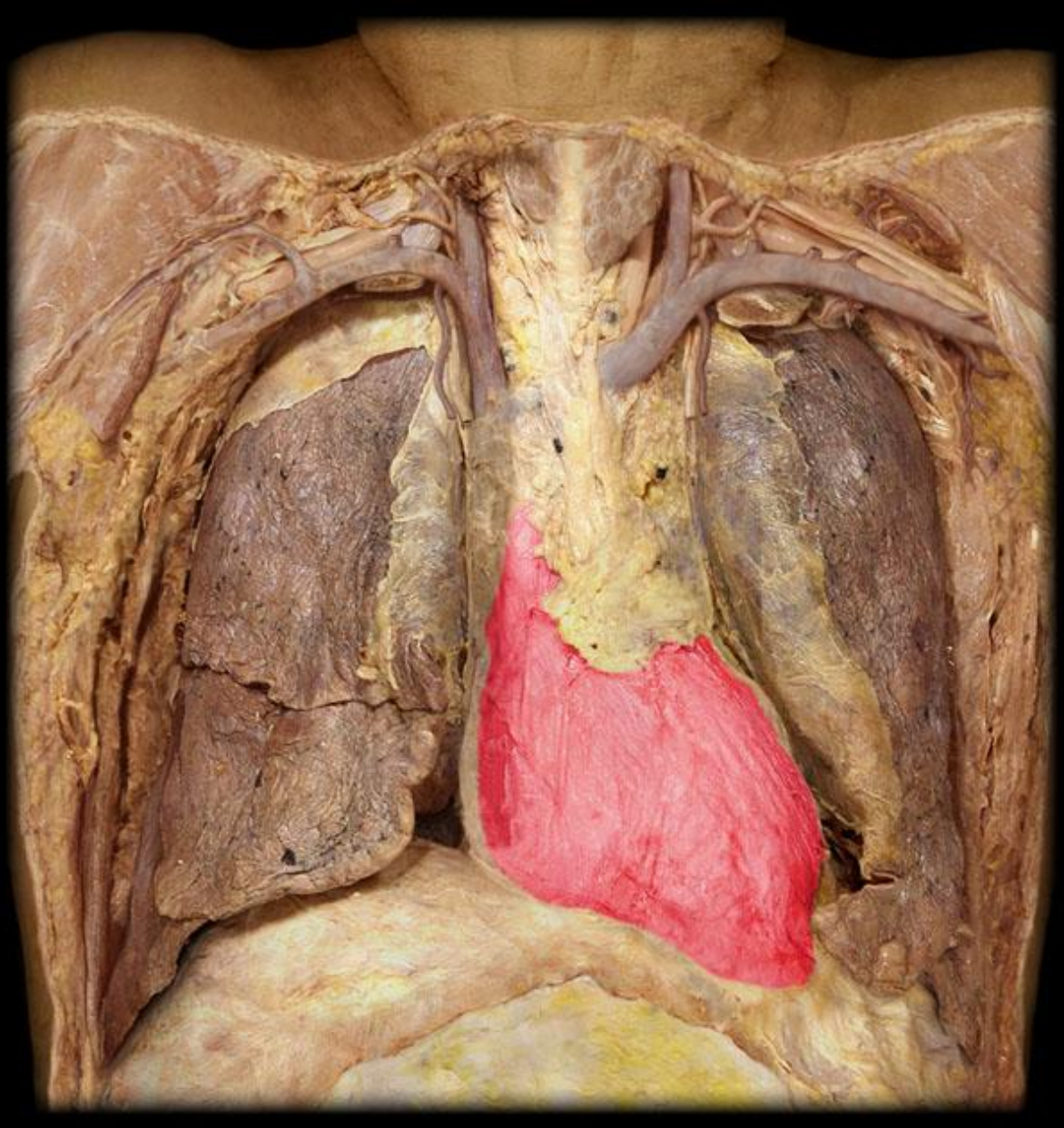
Lymphatic thymus



Final maturation of T-lymphocytes.

Blood-thymus barrier (special microenvironment for the maturing cells)

Adipose thymus



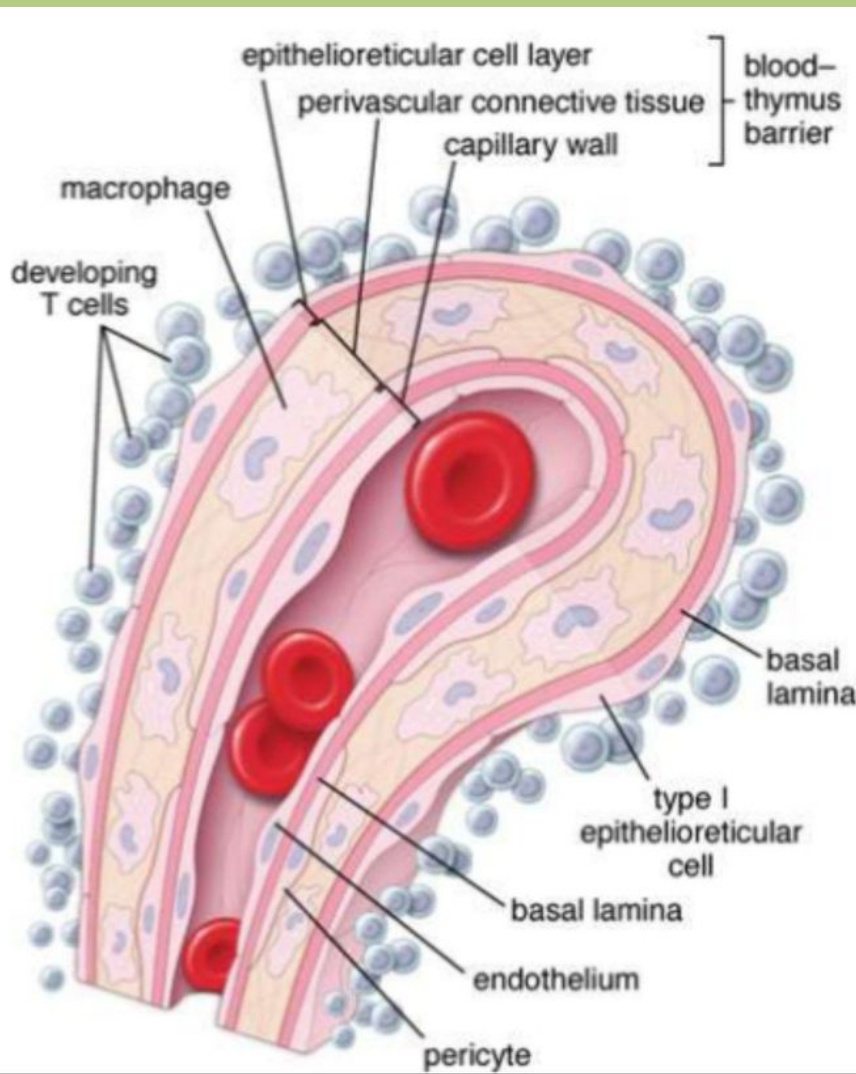
thymic involution
(until age 18-20)

Thymus

- two lobes
- CT capsule
- septae - lobules
- epithelioreticular (cytoreticular) meshwork
- NO lymphoid follicles
- cortex: thymocytes (T cells precursors)
- medulla: maturation of T cells
- Hassal's bodies: concentrically arranged layers of epithelial cells - keratohyalin and cytokeratin
- blood-thymus barrier

Blood-thymus barrier

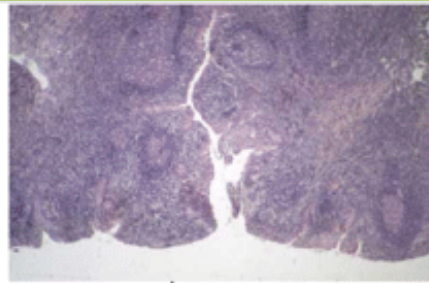
- special microenvironment in the cortex



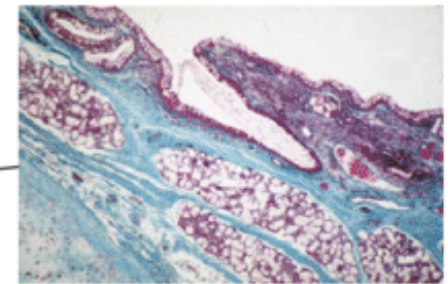
1. epithelioreticular cell layer
2. perivascular connective tissue with macrophages
3. basement membrane of endothelial cell
4. endothelial cells

Tonsils

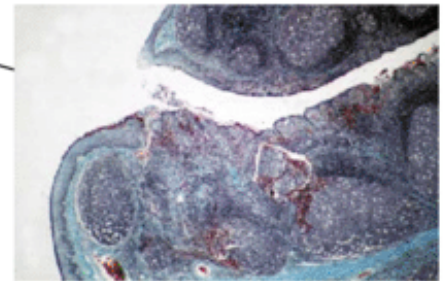
Waldeyer's lymphatic ring



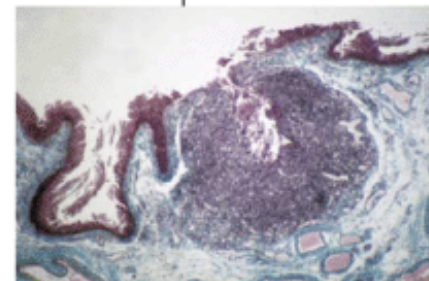
1 Adenoid = Pharyngeal tonsil



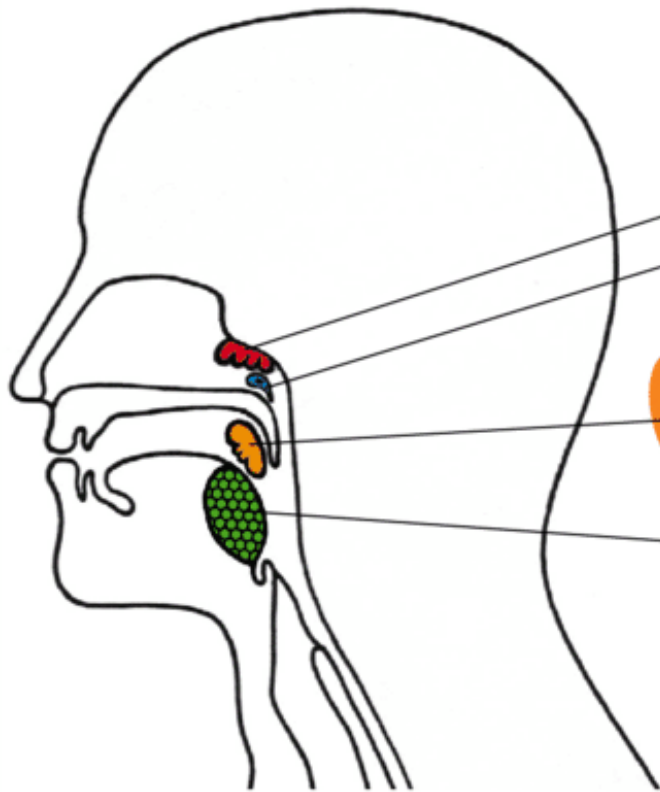
2 Tubal tonsil



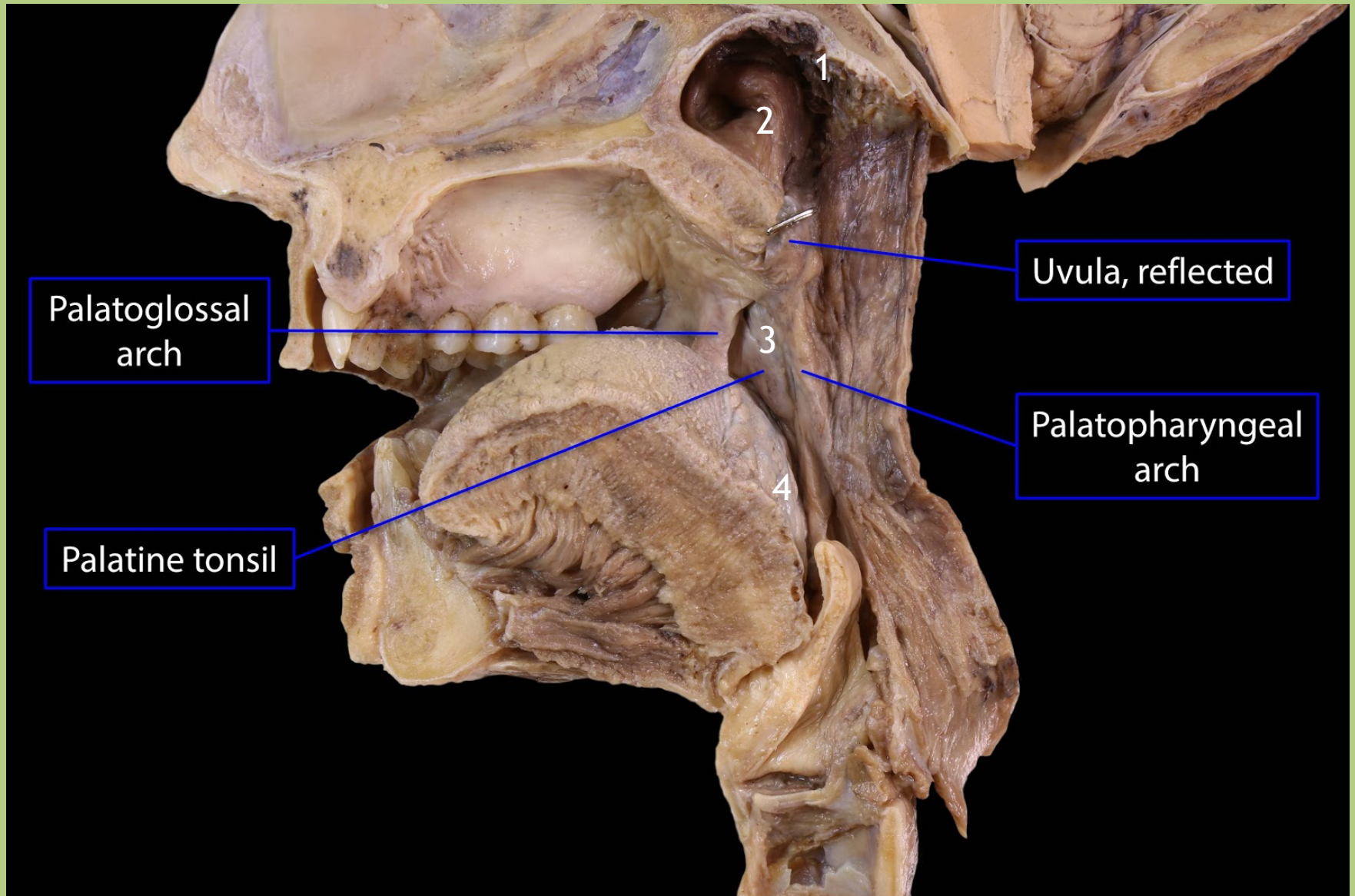
3 Palatine tonsil



4 Lingual tonsil



Waldeyer's lymphatic ring



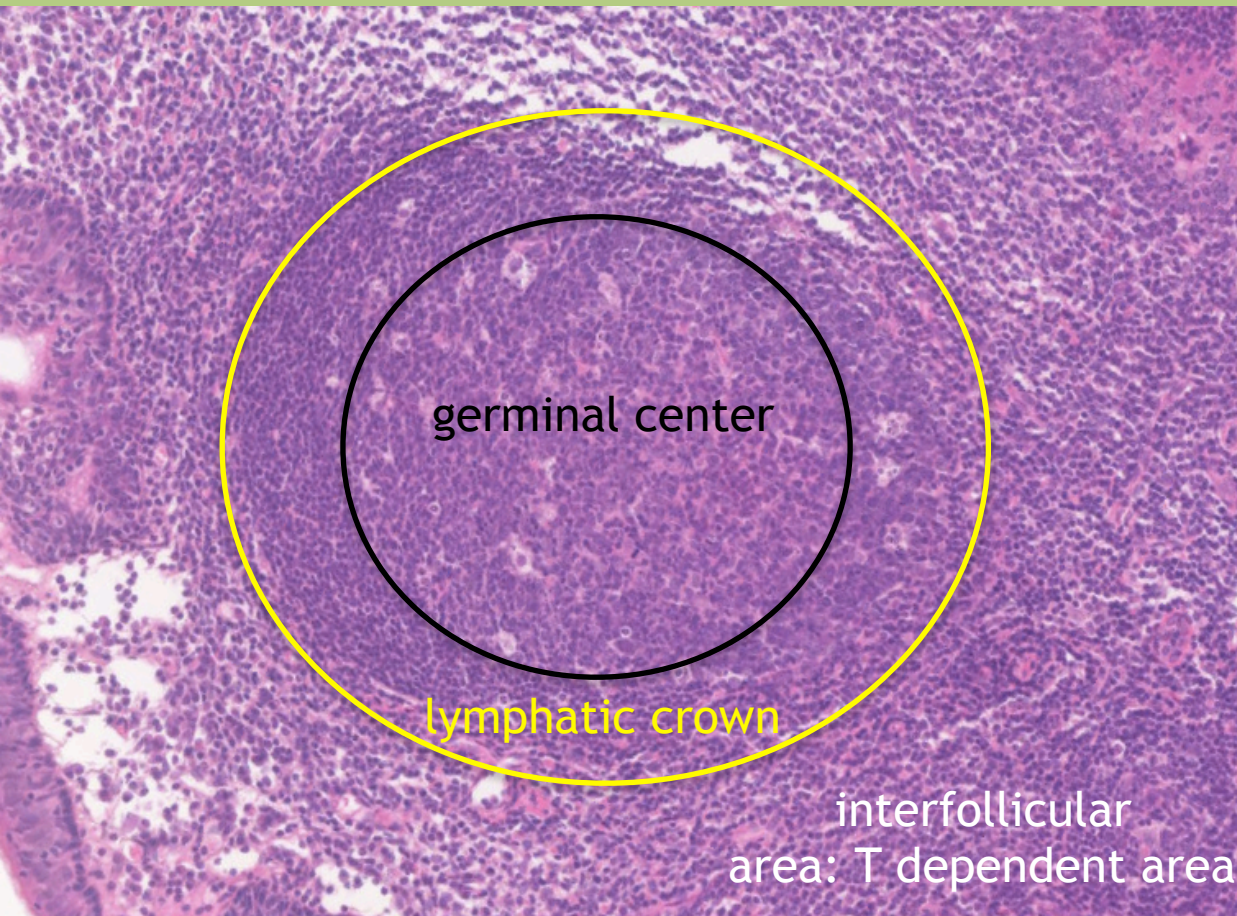
Hemisected head, medial

Pharyngeal tonsil=adenoid

- thin CT capsule
- pseudostratified columnar ciliated covering epithelium with goblet cells
- lymphoreticular meshwork
- secondary lymphoid follicles are presented (primer lymphoid follicles could be found in fetus)

Pharyngeal tonsil=adenoid

- secondary lymphoid follicles: B dependent areas



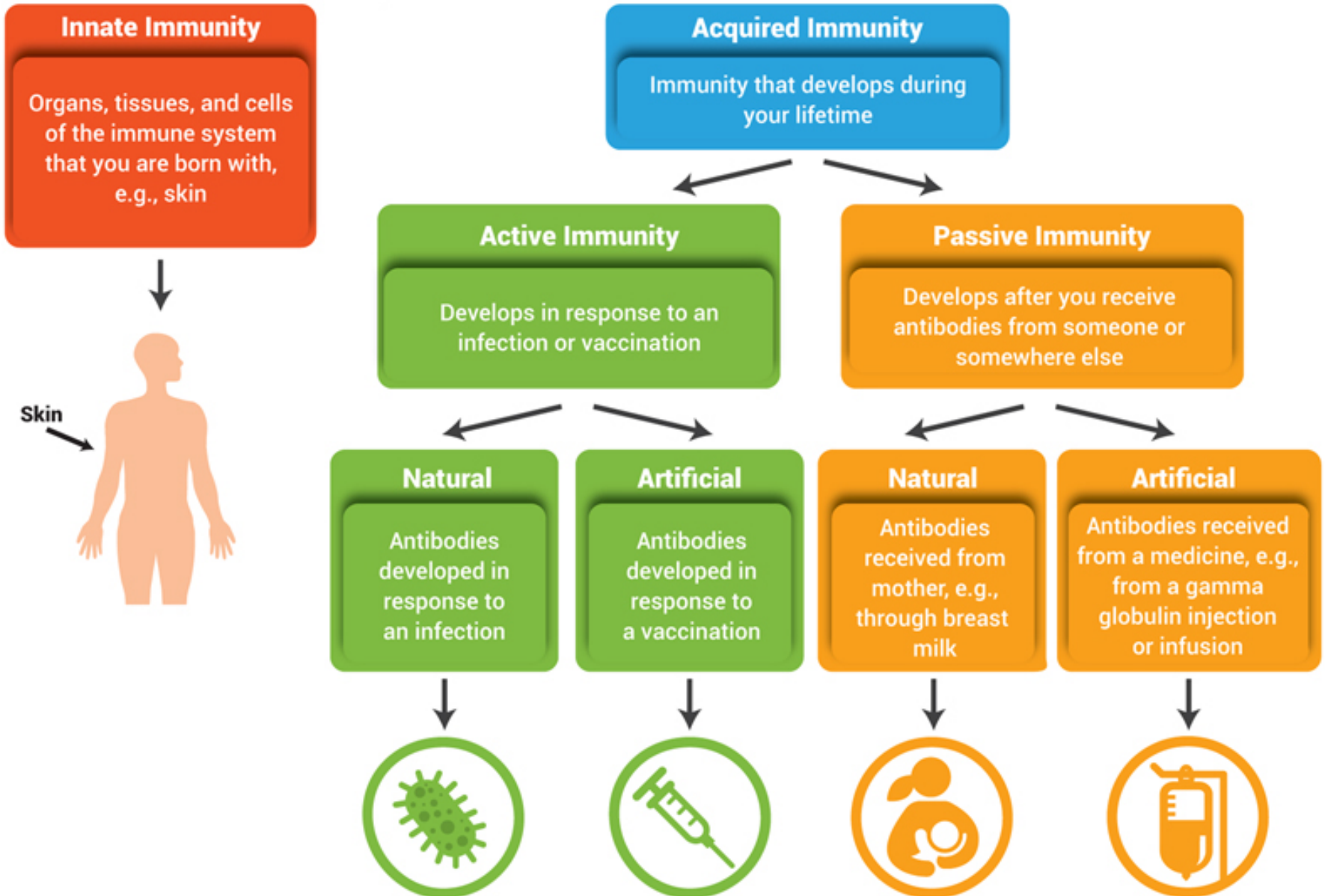
Palatine tonsil

- CT capsule
- stratified squamous non-keratinized covering epithelium
- lymphoreticular meshwork
- deep crypts
- secondary lymphoid follicles are presented (primer lymphoid follicles could be found in fetus)

Lingual tonsil

- NO capsule
- stratified squamous non-keratinized covering epithelium
- lymphoreticular meshwork
- shallow crypts
- secondary lymphoid follicles are presented (primer lymphoid follicles could be found in fetus)
- visceral striated muscle fibers and mucous glands of the tongue are also visible

Types of Immunity



Acquired Immunity

Immunity that develops during your lifetime

Active Immunity

Develops in response to an infection or vaccination

Natural

Antibodies developed in response to an infection



Artificial

Antibodies developed in response to a vaccination



Passive Immunity

Develops after you receive antibodies from someone or somewhere else

Natural

Antibodies received from mother, e.g., through breast milk



Artificial

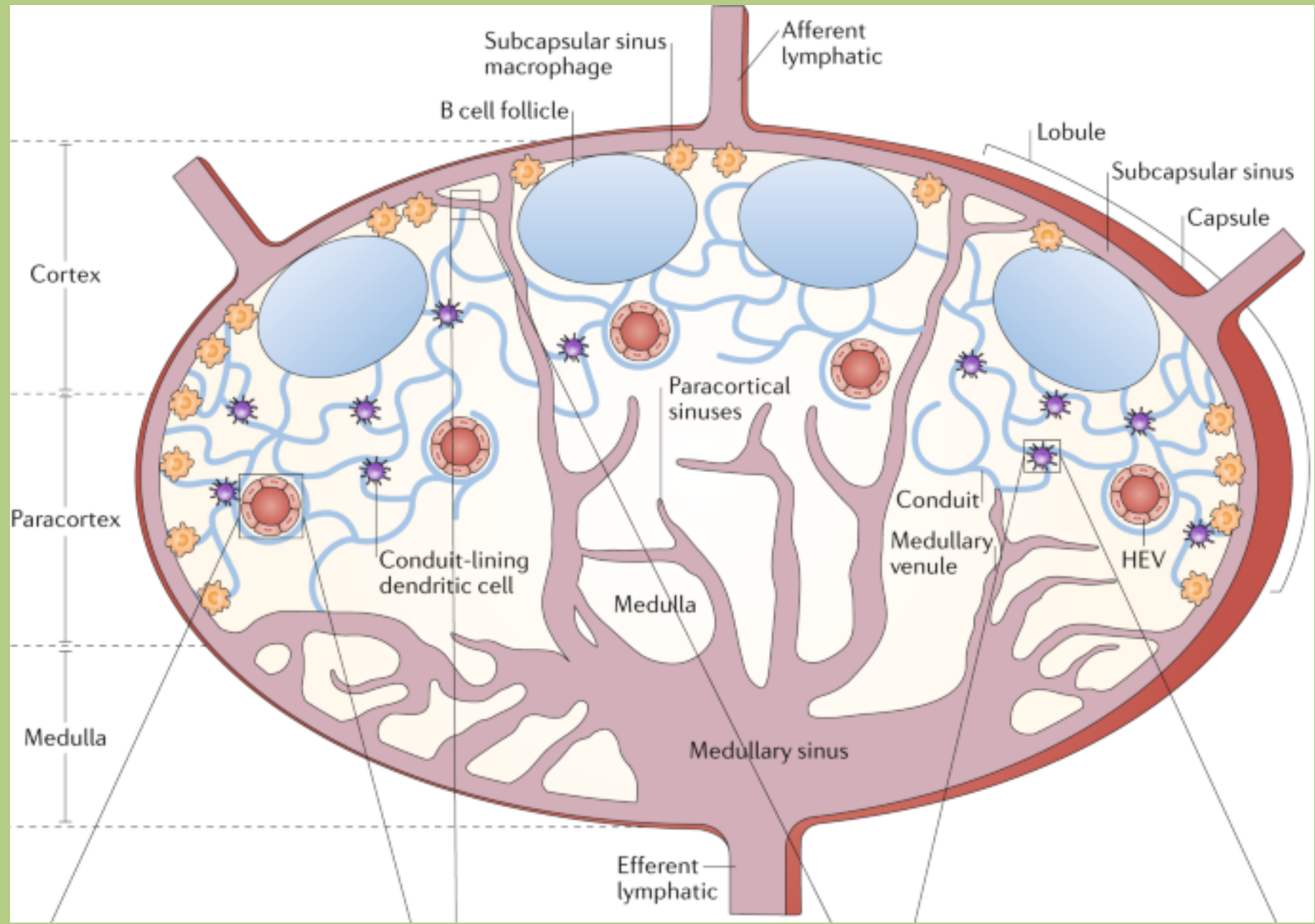
Antibodies received from a medicine, e.g., from a gamma globulin injection or infusion



Lymph node

- bean shaped
- 1 mm to 2 cm
- afferent lymph vessels-marginal sinuses-trabecular sinuses- medullary sinuses- efferent lymph vessel
- filters foreign particles
- lymphoreticular meshwork
- secondary lymphatic follicles in the cortex
- paracortex: no follicles
- medulla: B dependent area
- paracortex and interfollicular areas are T dependent areas
- HEVs in the paracortex for homing
- reticulum cells, tingible body macrophages, mast cells

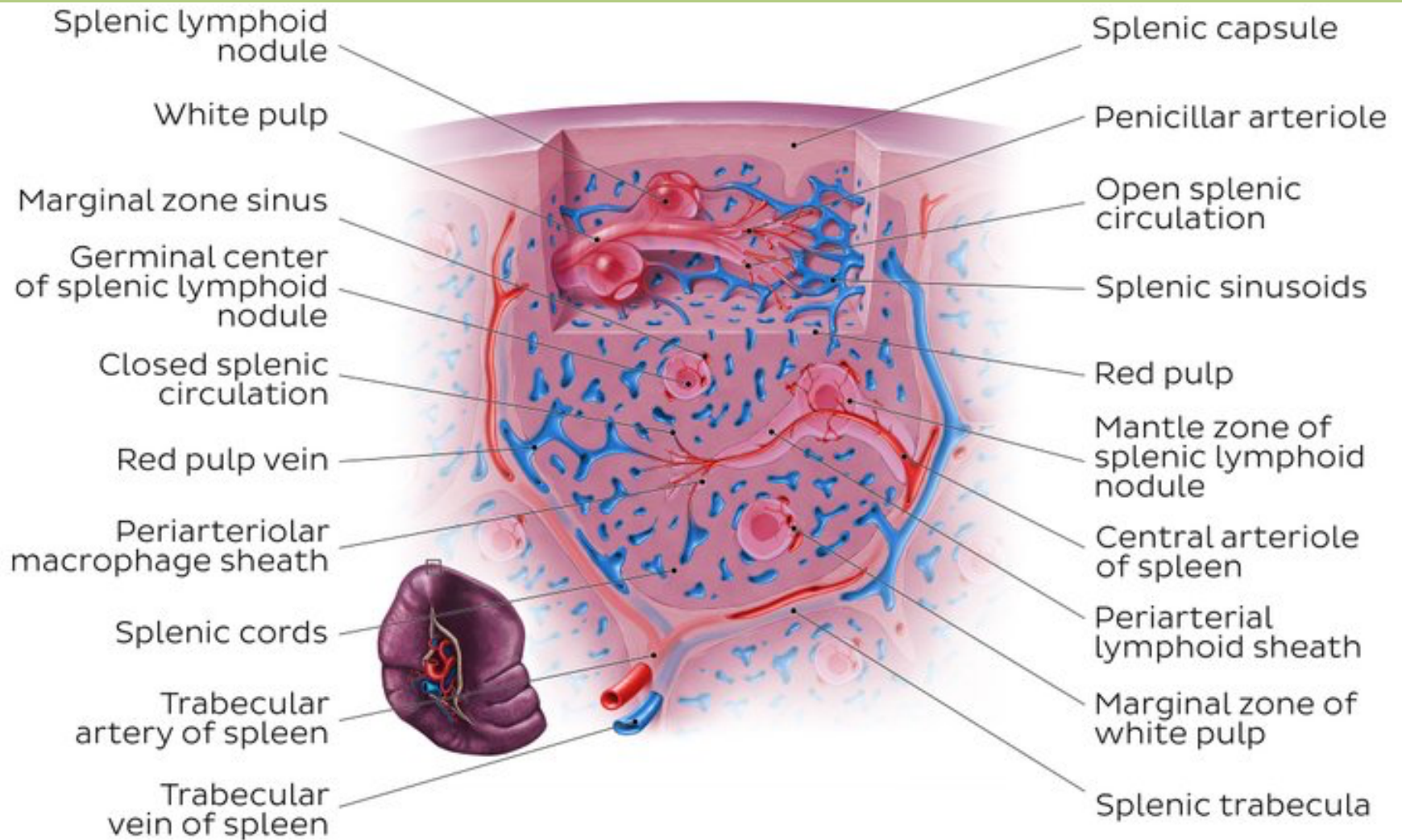
Lymph node



Spleen

- CT capsule
- trabeculae
- lymphoreticular meshwork
- red pulp:
 - Billroth's cords
 - splenic sinuses lined by stave cells
- white pulp:
 - Malpighian nodules (B dependent areas),
 - PALS (T dependent area),
 - marginal zones (B dependent areas)

Lymph node



Thank you for your attention.

References:

Pawlina: Histology, A Text and Atlas

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BlueLink.com